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Chief Technology Officer (CTO) | AI, SaaS & Enterprise Transformation Leader

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Engineering Automation & Digital Transformation Leader | Author of Mastering 3D Plant Engineering | SaaS Strategy | AI/ML | 28+ Years | India | SEA | Global

"I build digital bridges between engineering and business: simplifying real-world complexity into scalable systems aligned with strategy."

Executive Summary

Proven ability to scale platforms, teams, and revenue across global markets, delivering enterprise-grade business transformation. Strategic technology executive with 28+ years of experience driving enterprise-wide digital transformation across SaaS, manufacturing, and engineering domains. Proven ability to align technology vision with business strategy to deliver revenue growth, cost optimization, and scalable platform innovation.

Rare cross-domain leader who bridges **engineering, manufacturing, and SaaS platforms**, enabling practical, real-world solutions that directly impact business performance. Experience delivering global SaaS and enterprise platforms across India, SEA, US, and Europe. Operates at CTO/CIO/CDO level with strong ownership of product strategy, revenue enablement, and enterprise transformation.

Enterprise Value Creation & Revenue Impact

- Enabled shift from **SMB/mid-market to enterprise customer segment**, unlocking large global deals across US, Europe, and Japan
- Increased average customer renewal from **$2K–$5K → $35K+**, significantly improving ARR potential
- Introduced **consumption-based pricing model**, enabling profitability tracking per customer and improving deal conversion

- Built analytics-driven capabilities that **improved customer retention and product stickiness**
- Enabled enterprise-scale deployments (from **1K user limitation → unlimited scalability**)

Product & Technology Strategy Leadership

- Owned end-to-end product and technology strategy aligned with business growth objectives across AI, BI, NLP, SAM, and analytics platforms in alignment with CEO/founders
- Acted as key decision-maker on **feature prioritization based on customer demand and market direction**
- Worked closely with **sales, pre-sales, and customers** to shape product capabilities and close enterprise deals
- Positioned product as **AI-first, real-time analytics-driven platform**
- Enabled transition from traditional reporting to **NLP-based conversational analytics interface**

Board & Executive Leadership Engagement

- Worked directly with **CEO, founders, and owners** on technology and business strategy
- Presented **weekly roadmap, transformation progress, and strategic direction** to leadership
- Influenced major decisions across **product, expansion, hiring, and infrastructure investments**
- Played key role in **India and Moldova expansion strategy (infrastructure + team scaling)**
- Acted as **CTO-equivalent leader at Gunung**, contributing to public company transformation
- Led cross-functional enterprise domains including **AI, BI, ERP, SCM, Analytics,** and **Automation**

Core Competencies

- Technology Strategy & Business Alignment
- SaaS Architecture & Platform Engineering
- AI/ML & Real-Time Analytics Platforms
- Microservices & Cloud-Native Architecture
- Product Strategy & Roadmap Ownership

- Revenue Optimization & Pricing Strategy
- Industry 4.0 & Enterprise Digital Transformation
- Global Team Leadership & Scaling
- Stakeholder Management (Board, CXO, Investors)

Professional Experience

OpenLM, India

Tribe Lead (CTO Track Role – SaaS & AI Transformation)

Sept 2022 – Present

Strategic Mandate

Transform OpenLM into a scalable, AI-first SaaS platform aligned with enterprise market demands and global expansion.

Business & Technology Impact

SaaS Platform Transformation

- Re-architected legacy monolithic system into **microservices-based cloud platform**
- Built scalable SaaS platform supporting enterprise-grade deployments with distributed data architecture using Spark and Kafka
- Enabled multi-tenant architecture and global scalability
- Aligned product architecture with SaaS monetization strategy

Impact:

- Removed scalability limits (1K users → enterprise scale)
- Enabled entry into large enterprise market

Real-Time Data & Analytics Platform

- Built streaming analytics platform replacing batch-based reporting

Impact:

- Transitioned from **twice-daily reporting → real-time insights**

- Improved decision-making speed and customer experience

AI-Driven Engineering & DevOps

- Introduced AI-assisted development, testing, and CI/CD automation
- Implemented predictive alerting and anomaly detection for microservices.

Impact:

- Reduced development effort by 30–50%
- Increased release frequency from **1–2/year → daily deployments**

NLP-Based Product Innovation

- Developed NLP-based MCP reporting interface

Impact:

- Simplified complex BI into conversational queries
- Improved usability and customer adoption

Cost Optimization & Commercial Strategy

- Migrated to consumption-based licensing model
- Reduced Apache infra cost by 70%

Impact:

- Improved pricing flexibility and profitability tracking
- Reduced customer cost and improved competitiveness

System Performance & Reliability

- Implemented AI-based monitoring and anomaly detection
- Platform supports enterprise-scale deployments across global customers

Impact:

- Reduced downtime by 50%
- Reduced support tickets by 20%

Operational Efficiency

- Enabled instant customer onboarding (from 1 week → near real-time)
- Increased system capacity from GB to TB-scale data processing

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Organizational Scaling

- Scaled team from 3 → 80+ across engineering and support
- Achieved GEM certification for government contracts

BI Ecosystem Strategy

- Developed multi-platform BI architecture (Power BI, Superset, QuickSight).

Business Impact:

- Increased enterprise adoption through flexibility
- Enabled upsell and analytics-driven value creation

Gunung Raja Paksi, Indonesia

Head – Business Process (CTO-Equivalent Role)

Jan 2019 – Sept 2022

- Led digital transformation of large steel enterprise transitioning to public company
- Re-architected ERP across 25+ departments
- Transformed 50-year-old steel plant into a public company with modern digital backbone
- Built AI-driven procurement platform predicting scrap pricing and FX trends.
- Delivered enterprise BI dashboards for C-level executive decision-making.
- Enabled QR-code inventory tracking and mobile asset management
- Introduced analytics-driven procurement intelligence
- Delivered dashboards for real-time scrap cost optimization and location-based sourcing

Impact:

- ~$1M+ annual savings
- Reduced procurement cycle from **months → weeks → days**
- Improved inventory movement by 80%
- Enabled just-in-time material availability
- Improved operational transparency and governance
- Enabled data-driven strategic decisions at board level

Tata Steel, India

Senior Manager – Industry 4.0 & Virtual Engineering

Jan 2015 – December 2018

- Introduced drone, LiDAR, BIM, AR/VR and digital twin systems (industry-first).
- Pioneered Tata Steel's first-ever drone-based 3D modelling system for brown field plant engineering, production capacity expansion and maintenance: reducing planning time and enhancing asset visibility
- Built CAD automation tools using AutoLISP/VisualLISP for rapid drafting

Impact:

- Recovered ~30% revenue leakage
- Reduced engineering rework by 20%

M. N. Dastur & Co.

Tech Lead

Jan 2005 – January 2015

- Digitized legacy engineering workflows using BIM and automation
- Built enterprise document and CAD management systems.
- Cut engineering delivery cycle by 30% via design automation
- Spearheaded enterprise-wide GIS/CAD deployments
- Designed training programs adopted company-wide

Impact:

- Reduced workforce dependency by 68%
- Improved delivery efficiency and turnaround time

Cadworld Infosystems, India

R&D – CAD/GIS Automation

Apr 2000 – Mar 2005

- Led photogrammetry team contributing to Google 3D Maps.
- Built automation tools for GIS and CAD systems.
- Developed AutoLISP and DCL automation tools for civil CAD
- Delivered 3D GIS mapping services to global infrastructure firms
- Oversaw entire geospatial production chain:
    - Flight path design
    - Aero triangulation
    - Vector data capture
    - 3D modelling
    - Orthophoto generation

Business Impact:

- Improved productivity by 50%
- Delivered large-scale geospatial projects globally

Early Career

1. **RITES Ltd.** (1998 – 2000)– Automation of irrigation systems

    - Engaged in automation of irrigation gate opening
    - Automated plotting and feature extraction systems using Digital Theodolite data, reducing manual effort by 50%
    - Trained 50+ engineers in CAD automation techniques

2. **Larsen & Toubro** (1997 – 1998) – EPC engineering execution

    - Supported EPC execution at India's largest greenfield refinery, focusing on concrete structure alignment and layout precision

Cross-Domain Leadership Advantage

- Combines **engineering, manufacturing, and SaaS expertise** to design practical, business-aligned solutions
- Deep understanding of **customer problems from real industry experience (20+ years)**
- Ability to translate **business needs → product features → scalable systems**
- Bridges gap between **shop-floor operations and enterprise software platforms**

Innovation & Leadership Style

- Vision: To build **AI-first**, **real-time**, **scalable** and **cost-efficient** platforms that bridge business operations with intelligent decision-making systems.
- Challenges legacy systems and drives modernization
- Focuses on **practical execution over theoretical design**
- Balances **innovation with cost optimization**
- Designs systems that are **faster, scalable, easier, and more efficient than competitors**

Technology Expertise

Architecture & Platforms

- Microservices, Event-Driven Systems, SaaS Platforms

Data & AI

- Apache Spark, Kafka, Airflow, NLP, Predictive Analytics, Claude, Gemini, GPT

Cloud & DevOps

- AWS, Azure, Kubernetes, Docker, CI/CD, Swarm

Enterprise Systems

- SAP, Oracle, RPA (UiPath, Power Automate

Education & Certifications

- Master's in Computer Applications
- PG Diploma in IT
- Diplomas in Civil Engineering & Transport Management
- Six Sigma Green Belt | PMP
- AI/ML, BI, RPA Certifications

Positioning Statement

A CTO-level leader who combines deep technical expertise with strong business acumen, delivering scalable platforms, revenue growth, and enterprise transformation across industries.

Thought Leadership

Author of *Mastering 3D Plant Engineering, Digitalisation & Automation*

- Focus on engineering digitization, BIM, and automation strategy
- Widely used by industry professionals and transformation leaders

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Leadership Philosophy

- Align technology investments with measurable business outcomes
- Build scalable, future-ready platforms
- Drive innovation with execution discipline
- Bridge gap between business, operations, and technology

Publications & Authorship

Author of Mastering 3D Plant Engineering, Digitalisation & Automation — a field guide that blends engineering fundamentals, digital workflows, and automation insights.

🔷 Covers 3D modelling, drone scanning, BIM, and automation strategy
🔷 Used by plant engineering professionals, digitization consultants, and CIOs
📘 Amazon: B0F7M239VJ | 🌐 3dplantengineering.com | 🌐 soumenroy.com/publications

Professional Traits

- Strategically driven and execution focused
- Strong communicator with cross-functional leadership
- Ethical and transparent team mentor
- Results-oriented innovation mindset

Leadership Style

- Builds empowered, outcome-driven global teams
- Provides structure while encouraging agile innovation
- Creates strong alignment between product vision and execution
- Stands out by delivering measurable impact across tech-function-business interface

Significant Highlights

- 📈 Delivered 30% cost reduction in cloud infrastructure via architecture redesign
- 🛒 Launched predictive procurement system that forecasted scrap price & FX, saving ~$1M annually
- ✈️ Spearheaded drone-based 3D modelling at Tata Steel for the first time in industry history
- 🛰️ Introduced Satellite Based imagery and Aerial Photography for 3D modelling first time in world history with Google
- 🚀 Scaled OpenLM engineering teams from 3 to 85 across 4 product lines
- 🔄 Built real-time BI infrastructure using Spark, Airflow & Kafka
- 📚 Authored Mastering 3D Plant Engineering, Digitalisation & Automation, a first of-its-kind applied guide

Notable Projects

- **AI-driven scrap procurement optimization**: global supplier analysis with predictive ML dashboards
- **Drone-based 3D engineering modelling** across Tata Steel refining sites (> 800 acres)

- **Real-time streaming analytics** pipelines for SaaS dashboards in OpenLM
- **GIS/BIM-driven rework reduction** and faster engineering decision-making at Dastur & Tata
- **Digital twin & photogrammetry integration** for Google 3D Maps (Cadworld photogrammetry project)

Geographic Experience

- 🇮🇳 **India**: OpenLM, Tata Steel, M. N. Dastur, Cadworld
- 🇮🇩 **Indonesia**: Gunung Raja Paksi
- 🇨🇦 **Canada**: OSIG
- 🌐 **Global**: SaaS, analytics, and engineering delivery

Language

- English
- Hindi
- Bengali